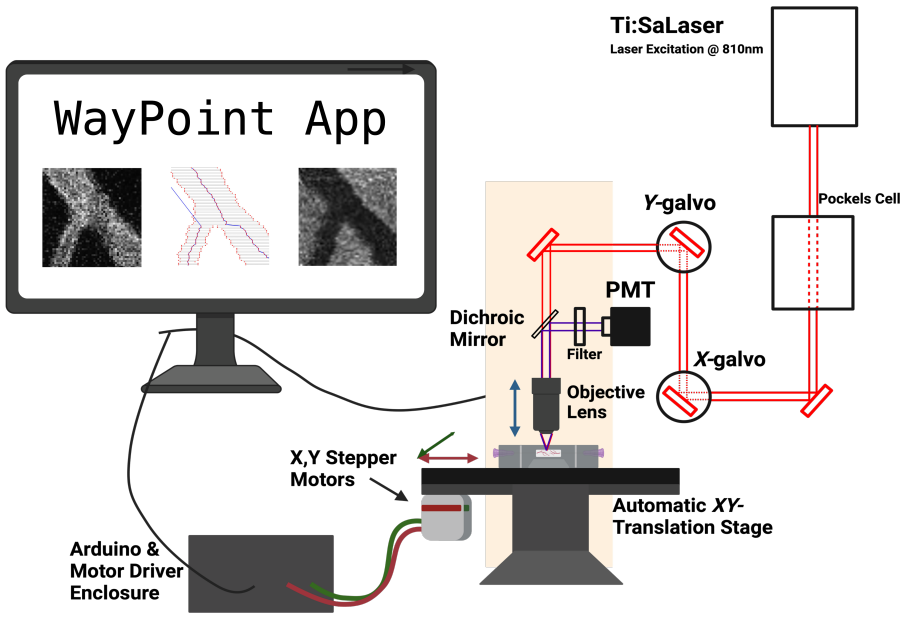
MAX HARKINS **mharkin4@u.rochester.edu**

WayPoint

December 2024 – March 2026
github.com/maxharkins/WPv3

About: MATLAB-based app that facilitates arbitrary 3D laser ablation on two-photon microscope.



Main contributions:

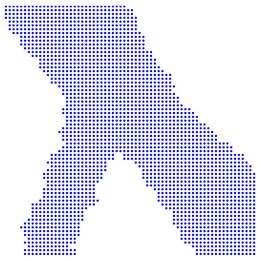
Full-stack app development

- Hardware-software integration
- Scan path generation
- X/Y galvo waveform control

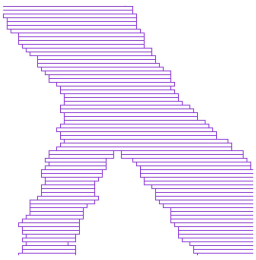
Built automatic XY stage

- Coupled steppers to linear stages
- Low-voltage stepper drive
- Serial PC-Arduino motor control

Scan path generation

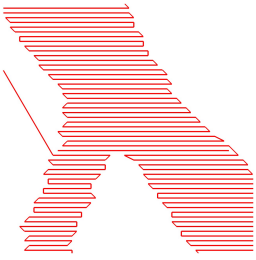


1 Binary mask



2 Horizontal runs

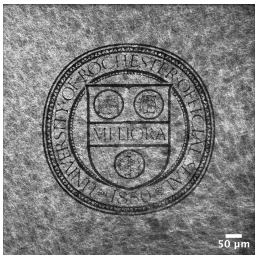
3 Midpoint traversal



4 Connected path

Fabricated structures in collagen

Image-derived vascular network



University of Rochester seal

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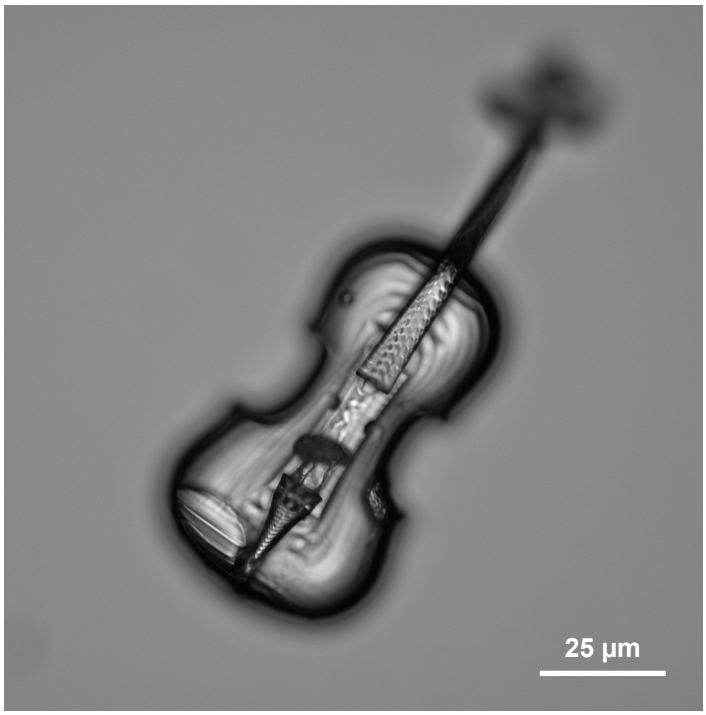
Resin Microfabrication

September 2026 – Present

About: Two-photon microprinting on the WayPoint system I built for laser ablation.

Violin STL model



Resin print under brightfield microscopy

Main contributions:

- Reused the WayPoint scan and focus controls to print the STL model layer by layer.
- Polymerized commodity 405 nm UV-curable resin using two-photon excitation at 810 nm.
- Washed away uncured resin with ethanol, then mounted and imaged the print.

Future directions:

- Design and fabricate integrated optical systems that need only initial input-beam alignment.
- Build my intuition by rapidly prototyping and testing microlens systems.

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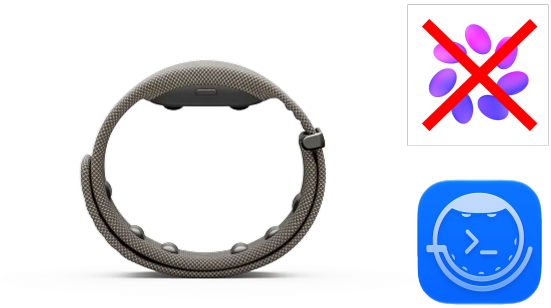
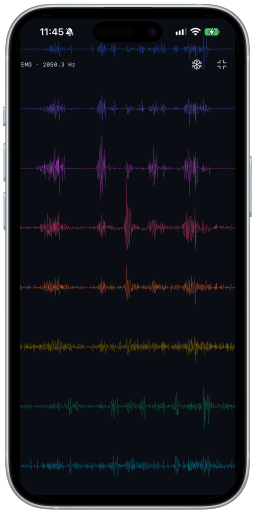
Meta Neural Band EMG

August – October 2026

Sedna: App for EMG streaming from the Meta Neural Band.
EMG Hand Kinematics Model: Predicts 22 joint angles from EMG.



Live EMG from Meta Neural Band



Main contributions:

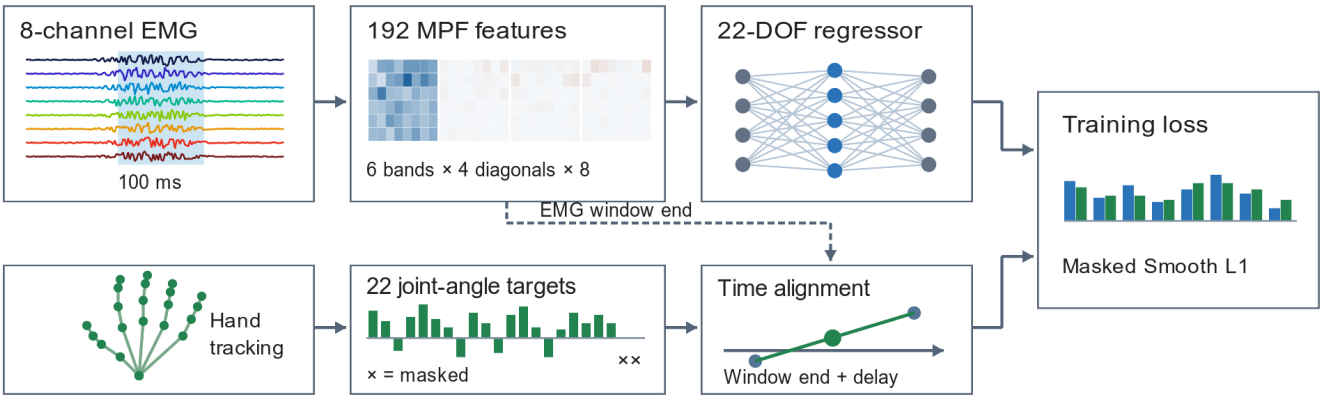
- Reverse engineered the Meta Neural Band protocol over BLE L2CAP CoC.
- Developed Sedna for iOS, Android, and macOS.
- Developed an MLP (inspired by EMG research at CMU) to predict hand joint angles.
- Integrated MediaPipe to label live EMG for online supervised training.

Hand kinematics

Camera with MediaPipe landmarks

22-DOF anatomical reconstruction

Model architecture and MediaPipe training labels



Regressor: LayerNorm(192) → Linear(192, 128) → ReLU → Linear(128, 22)